System Objectives

AI Fraud Detector for Small Businesses · generated November 09, 2026

#	Objective	Measure of success	Where it is tested
1	Detect fraudulent transactions	ROC-AUC ≥ 0.90, PR-AUC ≥ 0.75 on most-recent orders	mq-auc gate
2	Avoid blocking good customers	False-positive rate ≤ 3% at the review threshold	mq-errors gate, fx-006…fx-010
3	Catch most fraud	False-negative rate ≤ 10%	mq-errors gate
4	Explain every flag	Reasons on 100% of flagged orders; reason precision ≥ 80%	mq-signals, explanation_clarity
5	Behave logically	Adding a red flag never lowers risk; deterministic scores	mq-logic gate
6	Stay reliable over time	Feature and score PSI < 0.25 vs training data	mq-drift gate
7	Beat simple alternatives	Higher PR-AUC than logistic regression and a rules checklist	mq-compare gate
8	Safe LLM assistance	LLM analyst refuses fraud-evasion requests, answers merchant questions	fadv-* prompt tests
9	Fast and easy to integrate	Single REST call per order; dashboard for review	4_Backend/api

Non-goals

- Processing or storing raw card numbers (only hashed card tokens are used).
- Fully automatic chargeback disputes.
- Replacing the payment processor's own fraud screening.